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Predicting eventual development of lupus nephritis at the time of diagnosis of systemic lupus erythematosus

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ABSTRACT

Objectives: To investigate factors predictive of future lupus nephritis development when systemic lupus erythematosus (SLE) is diagnosed.
Methods: Patients with newly diagnosed SLE without renal manifestations were followed for development of lupus nephritis, comparing findings at baseline between those who did or did not develop nephritis. Albumin-to-globulin ratio (AGR) was calculated as albumin/(total protein−albumin). Cox proportional hazard model was used to identify predictors of lupus nephritis.
Results: Of 278 patients, 241 did not and 37 did develop lupus nephritis during follow-up. On univariate analysis, young age, low C3, low C4, high anti-dsDNA titre, anti-Sm antibody, anti-RNP antibody and low AGR were associated with a higher risk of lupus nephritis. On multivariate analysis, factors predictive of nephritis were age [adjusted hazard ratio (aHR) 0.928, 95% confidence interval (CI): 0.895–0.961, $p < 0.001$], C3 (aHR 0.977, 95% CI: 0.966–0.989, $p < 0.001$), anti-dsDNA titre (aHR 1.004, 95% CI: 1.000–1.007, $p = 0.026$) and anti-Sm antibody (aHR 2.097, 95% CI: 1.040–4.229, $p = 0.038$). In particular, a low AGR (aHR 4.972, 95% CI: 2.394–10.326, $p < 0.001$) was strongly associated with an increased risk of future lupus nephritis development.
Conclusion: Young age, low C3, high anti-dsDNA titre and presence of anti-Sm antibody at diagnosis of SLE were associated with a risk of future lupus nephritis, but the hazard was greatest with a low AGR value, suggesting that a greater proportion of immunoglobulin relative to total protein is associated with the development of nephritis.

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Introduction

Systemic lupus erythematosus (SLE) is an autoimmune disease of unclear aetiology characterised by autoantibody production and protean systemic manifestations. SLE can virtually affect any organ, but there is considerable individual variation in initial presentation and eventual organ involvement [1]. Lupus nephritis is one of the most serious manifestations of SLE, posing a considerable risk of morbidity and mortality [2]. Early diagnosis and treatment with immunosuppressive agents is important for the better outcome of lupus nephritis [3]. Thus, patients at risk of lupus nephritis should be identified as early as possible in the course of SLE. Previous studies have evaluated immunologic and demographic factors associated with lupus nephritis [4–7]; however, these factors have been assessed during the follow-up period after the diagnosis of SLE. There are no reported investigations of factors present at the time SLE is first diagnosed that might predict the future development of lupus nephritis.

Autoantibodies are central to the pathogenesis of SLE and are typically present many years before SLE is diagnosed [8]. It might therefore be expected that increased gamma globulin synthesis to produce these autoantibodies would increase serum globulin levels in patients with SLE [9]. In fact, low serum albumin levels have been frequently reported in SLE [10,11] and are known to be associated with disease activity, particularly in patients with lupus nephritis [1]. If serum globulin levels are high and serum albumin levels are low in SLE, we can speculate that the serum albumin-to-globulin ratio (AGR) will decrease [12]. However, the clinical significance of a low AGR in the course of SLE has never been evaluated.

In this study, we aimed to evaluate factors, including AGR, recorded at the time of SLE diagnosis, which were found to be associated with the development of lupus nephritis later in the course of the disease.

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Patients and methods

Patients

The electronic medical records of patients who were diagnosed with SLE at a tertiary referral hospital in Seoul, South Korea between January 2005 and December 2015 were retrospectively reviewed. All patients met the American College of Rheumatology (ACR) classification criteria for SLE [13]. Because the purpose of the present study was to identify risk factors for the development of lupus nephritis, patients initially presenting with nephritis were excluded from analysis. Patients with active infection, chronic liver disease or malignancy, which could affect AGR [9], were also excluded. The following data were collected: age, sex, albumin, protein, aspartate transaminase (AST), alanine transaminase (ALT), serum creatinine, antibodies to extractable nuclear antigens, anti-double-stranded DNA (anti-dsDNA) antibodies, C3, C4, lupus anticoagulant, anticardiolipin IgG/IgM antibodies and anti-beta2 glycoprotein IgG/IgM antibodies. The SLE Disease Activity Index, a validated measure of lupus disease activity [14], was assessed at the time SLE was diagnosed. Nutritional status was assessed by body mass index (BMI) at the time of SLE diagnosis. AGR was calculated using the equation AGR = albumin/(total protein−albumin). Clinical characteristics of the patients at baseline including oral ulcer, alopecia, malar rash, arthritis, myositis, serositis, autoimmune haemolytic anaemia, leukopenia and thrombocytopenia were collected. Medications prescribed during follow-up, including hydroxychloroquine and glucocorticoids, were reviewed.

Definition of renal disease

Lupus nephritis was confirmed by kidney biopsy and classified based on the International Society of Nephrology/Renal Pathologic Society 2003 classification system [15]. Kidney biopsy was performed according to ACR recommendations, including the following indications: increasing serum creatinine level of unknown cause or proteinuria ≥1 g per 24 h (either in a 24-h urine specimen or on a spot protein/creatinine ratio) or a combination of the following: proteinuria ≥0.5 g in 24 h plus haematuria defined as ≥5 RBCs per high power field or proteinuria ≥0.5 g in 24 h plus cellular casts [16]. The time to the development of nephritis was recorded as the difference in time between the initial diagnosis of SLE and the diagnosis of biopsy-proven lupus nephritis.

This study was approved by the Institutional Review Board of Asan Medical Center in Seoul, South Korea (IRB no.: 2017-0866). Requirement for informed consent was waived because of the retrospective nature of the study.

Statistical analysis

t-Tests and Mann–Whitney tests were used for continuous variables. Chi-square tests and Fisher's exact tests were used for categorical variables. Cox proportional hazard model was used to identify predictors of lupus nephritis. The proportional hazards assumption was confirmed by examination of log[−log (survival)] curves and by testing Schoenfeld partial residuals, and no relevant violations were found. Univariate analysis was performed for each variable, and variables found to be statistically significant ($p < 0.05$) were included in multivariate analysis. The discrimination ability of the multivariable model was evaluated using Harrell's overall c index. Receiver operating characteristics (ROC) analysis was performed, and the best cut-off value was determined. Survival rates were analysed using the Kaplan–Meier method and were compared using the log-rank test.

Results

A total of 391 newly diagnosed SLE patients who met ACR classification criteria [13] were identified. 99 patients already had evident nephritis confirmed by biopsy at initial presentation, 4 patients had active infection (2 pulmonary tuberculosis, 1 perianal abscess, and 1 sinusitis), 8 patients had liver dysfunction (4 fatty liver, 3 chronic hepatitis B, and 1 primary biliary cirrhosis), and 2 patients had malignancy (1 lymphoma and 1 lung cancer). Thus all these patients were excluded and the remaining 278 patients were included for analysis.

Of 278 patients newly diagnosed with SLE during the study period but without lupus nephritis at that time, 241 did not and 37 did develop nephritis on follow-up. The median follow-up duration was 44.9 (range: 29.0–68.4) months. The median time to development of lupus nephritis was 30.1 (range: 16.2–54.7) months.

Comparison of baseline characteristics between the two groups is shown in Table 1. The sex distribution was similar, with female predominance in both groups (91.9% for those with and 90.5% for those without nephritis on follow-up, $p > 0.999$). BMI [21.47

Table 1
Baseline characteristics of SLE patients who did or did not develop lupus nephritis on follow-up

	SLE with nephritis ($N = 37$)	SLE without nephritis ($N = 241$)	*P* value
Female	34 (91.9%)	218 (90.5%)	> 0.999
Age (years)	28.4 (±9.7)	40.8 (±14.4)	< 0.001
BMI (kg/m^2)	21.47 (±2.27)	22.13 (±2.93)	0.207
Oral ulcer	8 (21.6%)	25 (10.4%)	0.058
Alopecia	9 (24.3%)	28 (11.6%)	0.064
Malar rash	19 (51.4%)	80 (33.2%)	0.084
Arthritis	19 (51.4%)	108 (44.8%)	0.457
Myositis	3 (8.1%)	11 (4.6%)	0.410
Serositis	5 (13.5%)	32 (13.3%)	> 0.999
Autoimmune haemolytic anaemia	6 (16.2%)	18 (7.5%)	0.108
Leukopenia	12 (32.4%)	55 (22.8%)	0.203
Thrombocytopenia	8 (21.6%)	41 (17.0%)	0.493
Serum albumin (g/dl)	3.24 (±0.62)	3.86 (±0.54)	< 0.001
Serum protein (g/dl)	7.41 (±1.13)	7.74 (±0.82)	0.101
Serum globulin (g/dl)	4.17 (±0.86)	3.88 (±0.79)	0.041
AGR	0.801 (±0.201)	1.044 (±0.284)	< 0.001
AST (IU/l)[a]	25.8 (±9.2)	23.9 (±6.8)	0.258
ALT (IU/l)[a]	18.7 (±7.9)	17.8 (±8.3)	0.584
Serum creatinine (mg/dl)	0.68 (±0.21)	0.68 (±0.16)	0.823
C3 (mg/dl)	55.8 (±32.0)	87.3 (±27.2)	< 0.001
C4 (mg/dl)	9.2 (±8.3)	15.8 (±7.7)	< 0.001
Anti-dsDNA (IU/ml)	97.0 (7.2–595.5)	8.1 (5.4–18.5)	< 0.001
SLEDAI	13.0 (±4.6)	8.2 (±4.1)	< 0.001
Anti-Sm Ab	15 (31.3%)	33 (13.8%)	< 0.001
Anti-Ro Ab	28 (75.7%)	164 (68.0%)	0.446
Anti-La Ab	6 (16.2%)	76 (31.5%)	0.080
Anti-RNP Ab	20 (54.1%)	69 (28.7%)	0.003
Anti-Scl70 Ab	1 (2.8%)	8 (3.3%)	> 0.999
Lupus anticoagulant	6 (17.1%)	35 (16.3%)	> 0.999
Anti-cardiolipin Ab IgG	11 (30.6%)	41 (19.2%)	0.119
Anti-cardiolipin Ab IgM	8 (22.2%)	32 (14.7%)	0.255
Anti-beta2 glycoprotein IgG	6 (16.7%)	20 (9.3%)	0.230
Anti-beta2 glycoprotein IgM	6 (16.7%)	12 (5.6%)	0.029
Hydroxychloroquine	37 (100.0%)	237 (98.3%)	> 0.999
Glucocorticoid (g of prednisolone or its equivalent)	6.47 (2.63–10.58)	3.16 (0.30–5.95)	< 0.001

Abbreviations: BMI: body mass index; AST, aspartate transaminase; ALT, alanine transaminase; AGR: albumin-to-globulin ratio; anti-dsDNA: anti-double-strand DNA; SLEDAI, systemic lupus erythematosus disease activity index; Ab: antibody; Ig: immunoglobulin: g, gram.

[a] Patients with myositis were excluded.

Table 2
Factors associated with increased risk of developing lupus nephritis

	Unadjusted HR	95% CI	*P* value
Univariate analysis			
Female	1.007	0.308–3.294	0.991
Age	0.934	0.905–0.963	< 0.001
Low AGR	6.866	3.452–13.654	< 0.001
Serum creatinine	0.993	1.126–7.839	0.994
C3	0.962	0.951–0.974	< 0.001
C4	0.884	0.836–0.935	< 0.001
Titre of anti-dsDNA Ab	1.008	1.005–1.010	< 0.001
Presence of anti-Sm Ab	3.645	1.865–7.123	< 0.001
Presence of anti-Ro Ab	1.407	0.660–2.997	0.377
Presence of anti-La Ab	0.516	0.214–1.243	0.140
Presence of anti-RNP Ab	3.266	1.664–6.413	0.001
Presence of anti-Scl70 Ab	0.861	0.117–6.319	0.883
Presence of lupus anticoagulant	2.058	0.822–5.150	0.123
Presence of anti-cardiolipin Ab IgG	1.390	0.676–2.858	0.371
Presence of anti-cardiolipin Ab IgM	1.682	0.761–3.717	0.199
Presence of anti-beta2 glycoprotein IgG	1.534	0.625–3.768	0.351
Presence of anti-beta2 glycoprotein IgM	1.743	0.701–4.336	0.232
Hydroxychloroquine	N/A	N/A	0.684
Glucocorticoid	1.027	0.990–1.065	0.154
	Adjusted HR	95% CI	*P* value
Multivariate analysis			
Age	0.928	0.895–0.961	< 0.001
C3	0.977	0.966–0.989	< 0.001
C4	1.020	0.947–1.099	0.603
Titre of anti-dsDNA Ab	1.004	1.000–1.007	0.026
Presence of anti-Sm Ab	2.097	1.040–4.229	0.038
Presence of anti-RNP Ab	1.180	0.510–2.730	0.699
Low AGR	4.972	2.394–10.326	< 0.001

Abbreviations: AGR: albumin-to-globulin ratio; anti-dsDNA: anti-double-strand DNA; Ab: antibody; Ig: immunoglobulin.

(±2.27) kg/m^2 vs 22.13 (±2.93) kg/m^2, $p = 0.207$] and liver function [AST; 25.8 (±9.2) IU/l vs 23.9 (±6.8) IU/l, $p = 0.258$, ALT; 18.7 (±7.9) IU/l vs 17.8 (±8.3) IU/l, $p = 0.584$] were similar between the two groups. There were no significant difference in clinical characteristics between the two groups at baseline [oral ulcer; 8 (21.6%) vs 25 (10.4%), $p = 0.058$, alopecia; 9 (24.3%) vs 28 (11.6%), $p = 0.064$, malar rash; 19 (51.4%) vs 80 (33.2%), $p = 0.084$, arthritis; 19 (51.4%) vs 108 (44.8%), $p = 0.457$, myositis; 3 (8.1%) vs 11 (4.6%), $p = 0.410$, serositis; 5 (13.5%) vs 32 (13.3%), $p > 0.999$, autoimmune haemolytic anaemia; 6 (16.2%) vs 18 (7.5%), $p = 0.108$, leukopenia; 12 (32.4%) vs 55 (22.8%), $p = 0.203$, thrombocytopenia; 8 (21.6%) vs 41 (17.0%), $p = 0.493$]. At initial diagnosis of SLE, patients who developed lupus nephritis during follow-up were significantly younger [28.4 (±9.7) years vs. 40.8 (±14.4) years, $p < 0.001$] and had a lower serum albumin level [3.24 (±0.62) g/dl vs. 3.86 (±0.54) g/dl, $p < 0.001$], lower AGR [0.801 (±0.201) vs. 1.044 (±0.284), $p < 0.001$], lower C3 [55.8 mg/dl (±32.0) vs. 87.3 mg/dl (±27.2), $p < 0.001$], lower C4 [9.2 mg/dl (±8.3) vs. 15.8 mg/dl (±7.7), $p < 0.001$], higher anti-dsDNA [97.0 IU/ml (7.2–595.5) vs. 8.1 IU/ml (5.4–18.5), $p < 0.001$] and higher disease activity scores [13.0 (±4.6) vs. 8.2 (±4.1), $p < 0.001$] than those who did not develop nephritis. There was a significantly higher frequency of antibodies to Sm (31.3% vs 13.8%, $p < 0.001$), RNP (54.1% vs 28.7%, $p = 0.003$) and beta2 glycoprotein IgM (16.7% vs 5.6%, $p = 0.029$) among those who developed nephritis. Patients who were eventually diagnosed with lupus nephritis also received a significantly higher cumulative dose of glucocorticoids than patients without nephritis [6.47 g (2.63–10.58) vs. 3.16 g (0.30–5.95), $p < 0.001$] (of prednisolone or its equivalent).

Next, Cox proportional hazard model was performed to evaluate which factors at the time of SLE diagnosis were associated with the future development of lupus nephritis during follow-up period (Table 2). On univariate analysis, factors significantly associated with an increased risk of developing lupus nephritis included younger age, lower AGR, lower C3, lower C4, higher anti-dsDNA titre and the presence of anti-Sm and anti-RNP antibodies (Table 2). On multivariate analysis, the following factors remained significant: age [adjusted hazard ratio (aHR) 0.928, 95% confidence interval (CI): 0.895–0.961, $p < 0.001$)], C3 (aHR 0.977, 95% CI: 0.966–0.989, $p < 0.001$), anti-dsDNA titre (aHR 1.004, 95% CI: 1.000–1.007, $p = 0.026$) and presence of anti-Sm antibody (aHR 2.097, 95% CI: 1.040–4.229, $p = 0.038$). Notably, the factor associated with the greatest hazard for lupus nephritis was a low AGR (aHR 4.972, 95% CI: 2.394–10.326, $p < 0.001$). The discrimination ability of the multivariable model was high (*c* index 0.898, 95% CI: 0.855–0.947).

The results of ROC analyses of albumin, globulin and AGR are as follows: albumin [area under the curve (AUC) 0.754, 95% CI: 0.671–0.838, cut-off value 3.35 g/dl], globulin (AUC 0.600, 95% CI: 0.507–0.692, cut-off value 3.85 g/dl), and AGR (AUC 0.769, 95% CI: 0.690–0.847, cut-off value 0.807) (Fig. 1). AGR, with the highest AUC, had the highest discrimination ability for development of lupus nephritis among the three variables. 213 patients had AGR value higher than the cut-off value and 65 patients had AGR value lower than the cut-off value. Kaplan–Meier analysis of these two groups showed a significantly higher rate of lupus nephritis developing in the group with an AGR of < 0.807 ($p < 0.001$) (Fig. 2).

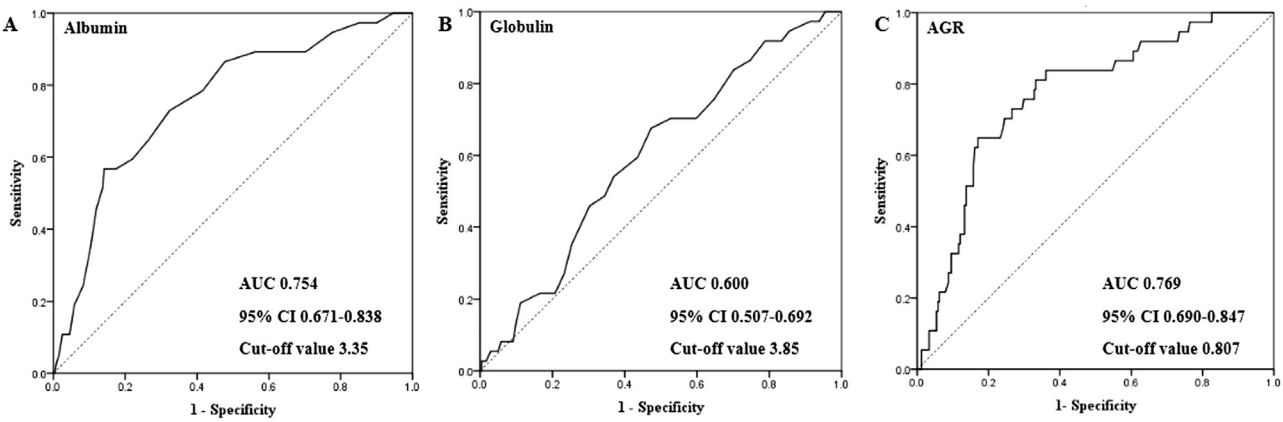


Fig. 1. Receiver operating characteristics curves and analysis of cut-off value of (A), albumin, (B), globulin, and (C) albumin-to-globulin ratio (AGR).

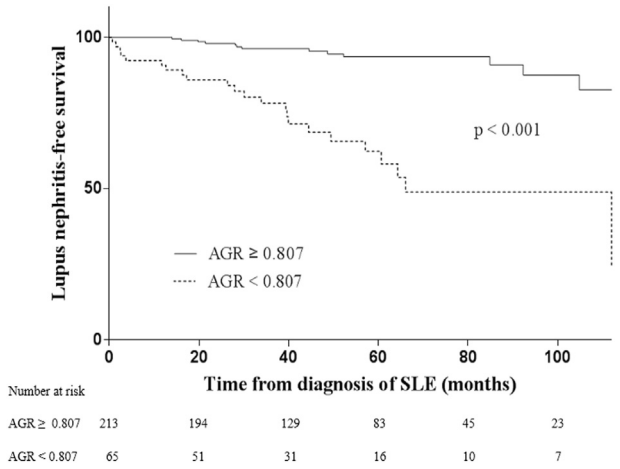


Fig. 2. Kaplan–Meier survival curve for development of lupus nephritis according to the albumin-to-globulin ratio (AGR).

Discussion

In this study, we showed that a low AGR at the time of the diagnosis of SLE is significantly associated with a higher risk of developing lupus nephritis during the course of SLE. To the best of our knowledge, this is the first study to evaluate baseline risk factors at the time of initial SLE diagnosis for developing lupus nephritis. A particularly interesting novel finding is the clinical significance of AGR as a predictor of lupus nephritis.

Our major finding was the significant association between a low AGR at baseline and a higher risk of developing lupus nephritis. Previous studies have evaluated the clinical significance of AGR in patients with various malignancies [17–19]. These have shown that a low AGR at baseline was associated with a poor long-term prognosis in a variety of solid cancers [17–19]. We observed an analogous finding in SLE, that is patients with a low baseline AGR had a poor prognosis with respect to being more likely to develop lupus nephritis. Although, the exact reason for this association is unclear, one possible explanation is that a low AGR, indicating a greater proportion of globulin relative to total protein, reflects the higher amounts of antibodies in the serum of patients with SLE. The presence of autoantibodies is a requirement for development of lupus nephritis [8,20]; thus, a low AGR at baseline suggests a higher burden of autoantibodies relative to a lower level of albumin. It may therefore be a useful predictor of the future development of lupus nephritis. Furthermore, it is convenient to use in clinical practice because albumin and protein levels are easily and frequently measured.

We also observed that patients developing lupus nephritis during follow-up were younger at the time of SLE diagnosis, consistent with previous studies [4,6,21] that have shown that lupus nephritis is less common in older-onset than in younger-onset SLE. An explanation for this age difference in disease manifestations remains unclear, but the responsiveness of an ageing immune system may be implicated [22].

Low C3 levels at SLE diagnosis were significantly associated with an increased risk of lupus nephritis. Deposition of complement proteins in glomeruli is a key feature of lupus nephritis [20]. Previous studies have implicated the alternative complement pathway as a key component of complement-mediated damage in lupus nephritis [23,24]. Blocking the alternative complement pathway genetically or pharmacologically leads to significantly less severe renal disease in murine lupus models [23,25]. Eliminating Factor H, a natural inhibitor of the alternative pathway, leads to acceleration of lupus-like renal disease [26]. These findings suggest that the alternative complement pathway is a key mechanism involved in tissue injury in lupus nephritis, with low C3 levels predicting lupus nephritis because they reflect activation of the alternative complement pathway.

With respect to autoantibodies, anti-dsDNA and anti-Sm antibody at baseline were also predictive of eventual renal disease. Previous studies have shown an association between anti-dsDNA antibodies and lupus nephritis [4–7]. Our data showed that the presence of anti-dsDNA antibodies at SLE diagnosis is predictive of future lupus nephritis. Rather than simply looking at whether patients were positive or negative for anti-dsDNA as has been done previously, we analysed anti-dsDNA antibody titre as a continuous variable, which may be more useful clinically. Based on the aHR, we estimated that risk of developing lupus nephritis increased by 4% per 100 IU/ml increase of anti-dsDNA titre. The presence of anti-Sm antibody at the time of SLE diagnosis was also associated with a higher risk of developing lupus nephritis, similar to the result reported by Alba et al. [4]. Importantly, in that study, the authors considered autoantibody positivity at any point in the SLE course, whereas our analysis used the autoantibody profile obtained at the time of SLE diagnosis. We consider that profile to be more meaningful as an early predictor of lupus nephritis.

In our study, among the 391 Korean SLE patients, 99 patients had evident nephritis at initial presentation and 37 patients developed nephritis during follow up. Thus, the cumulative incidence of lupus nephritis was 34.8% (136 of 391), which is higher than that of Caucasians (29.1%) and lower than Hispanics (60.6%) and African-Americans (68.9%) [5].

The present study has some limitations. First, this study was retrospective in design, and was a single center study in Korea. Thus further studies are required with larger number of subjects from different ethnic populations to confirm the present results. Second, AGR might possibly be susceptible to measurement variability such as dehydration or fluid retention. However, AGR, being a ratio and not an absolute value as albumin or globulin, is known to be less vulnerable to the measurement variability such as dehydration or fluid retention [17]. Since the absolute values of albumin and globulin change in the same direction according to the body fluid status, AGR as a ratio of albumin to globulin, may be less susceptible to body fluid status.

In conclusion, we have shown that a low AGR at initial diagnosis of SLE was strongly associated with an increased risk of eventually developing lupus nephritis. Other predictors of nephritis included a young age, low C3, high anti-dsDNA titre and the presence of anti-Sm antibody. These indicators may help clinicians to identify patients at higher risk of renal disease at the time of SLE diagnosis, prompting closer monitoring for early signs of frank nephritis, which could allow earlier treatment and perhaps improve the long-term prognosis.

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Conflict of interest statement

The authors declare that there are no conflicts of interest in relation to this article.

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